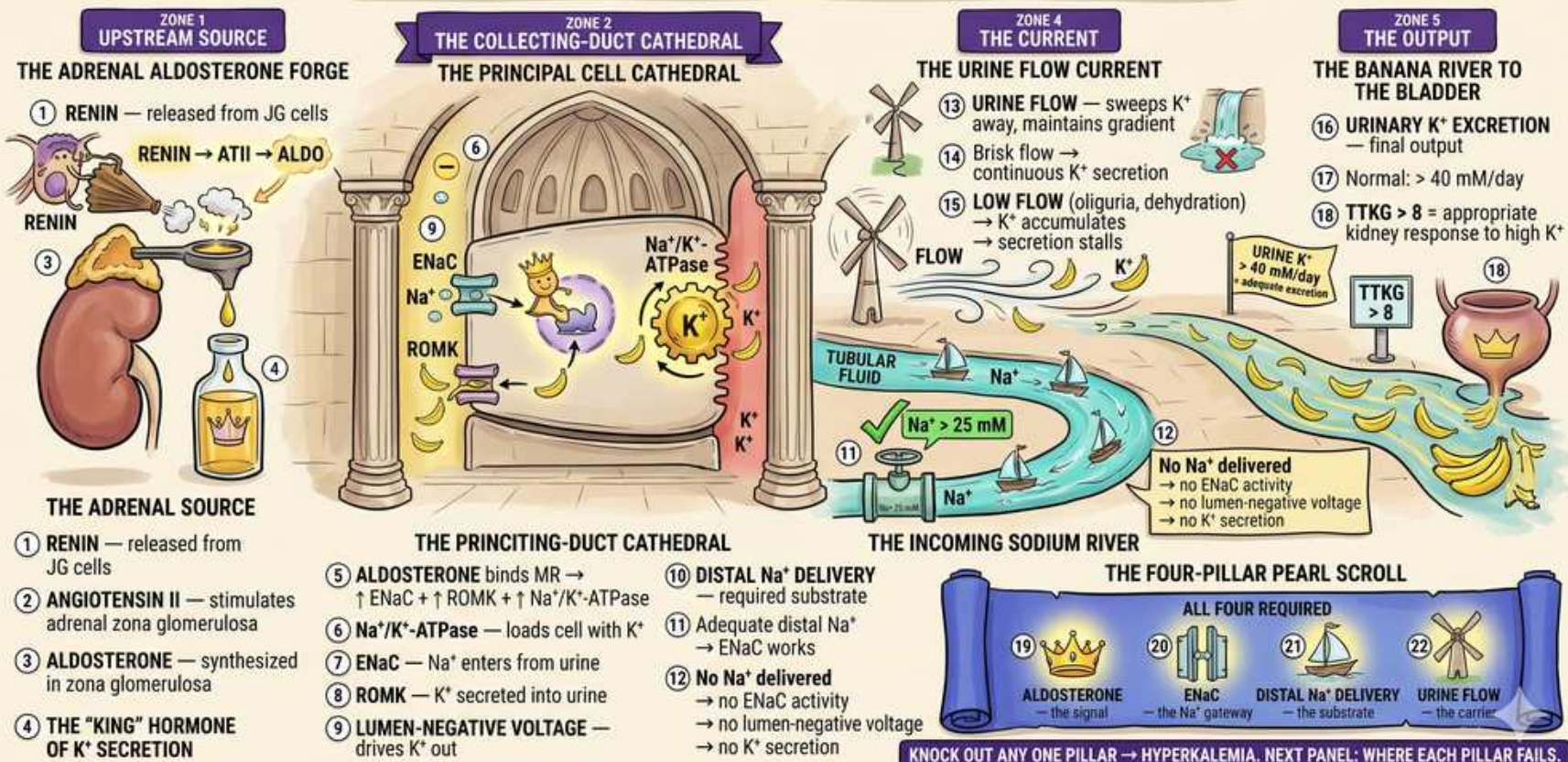


Hyperkalemia — nephron K⁺ secretion (4 pillars)

THE ROYAL KIDNEY EXCRETION PIPELINE — HOW THE NEPHRON SECRETES K⁺

K⁺ secretion needs **ALL FOUR**: Aldosterone + ENaC + Distal Na⁺ delivery + Urine flow



1. All four required

WHAT YOU SEE

Banner: K⁺ secretion needs ALL four

WHAT IT ENCODES

Distal nephron K⁺ secretion (principal cells of the late distal tubule/collecting duct) requires all four pillars: aldosterone effect, functioning ENaC, adequate distal Na⁺ delivery, and adequate urine flow. These are serial dependencies — each sets up the next, and the lumen-negative voltage from ENaC-mediated Na⁺ reabsorption is what drives K⁺ out through ROMK. Knocking out any one pillar impairs secretion and causes hyperkalemia.

BOARD TRAP

WRONG to assume normal serum aldosterone guarantees normal K⁺ excretion. Discriminator: even with intact aldosterone, K⁺ retention occurs if distal Na⁺ delivery is low (CHF, cirrhosis, volume depletion), flow is low (oliguria), or ENaC is blocked (amiloride/triamterene) — all four pillars must be intact.

2. Renin

WHAT YOU SEE

Renin released from JG cells

WHAT IT ENCODES

Renin, released from juxtaglomerular cells of the afferent arteriole, is the rate-limiting first step of the RAAS. It is stimulated by low renal perfusion pressure, low distal Na⁺ (macula densa), and sympathetic (beta-1) activity. Renin cleaves angiotensinogen to angiotensin I, initiating the cascade toward aldosterone.

BOARD TRAP

WRONG to overlook drugs that suppress renin as a cause of hyperkalemia. Discriminator: NSAIDs and beta-blockers lower renin (hyporeninemic states), reducing downstream aldosterone and K⁺ secretion — these are common, easily missed contributors in CKD/diabetic patients.

3. Angiotensin II

WHAT YOU SEE

ATII forge

WHAT IT ENCODES

ACE (mainly in pulmonary endothelium) converts angiotensin I to angiotensin II, which stimulates the adrenal zona glomerulosa to synthesize aldosterone and also drives proximal Na⁺/water reabsorption and efferent arteriolar constriction. It is the pivotal effector linking volume status to aldosterone. Blocking it impairs distal K⁺ secretion.

BOARD TRAP

WRONG ANSWER: failing to attribute hyperkalemia to ACE inhibitors/ARBs. Discriminator: ACEi (block AT-II formation) and ARBs (block AT-1 receptor) both reduce aldosterone-driven K⁺ secretion — they are leading drug causes of hyperkalemia, especially combined or in CKD/diabetes.

4. Aldosterone

WHAT YOU SEE

Aldosterone synthesized in zona glomerulosa

WHAT IT ENCODES

Aldosterone, made in the adrenal zona glomerulosa, is the master hormone for distal K⁺ secretion: it upregulates ENaC, the basolateral Na/K-ATPase, and ROMK in principal cells. It is stimulated directly by hyperkalemia and by angiotensin II. Deficiency (Addison's, hyporeninemic hypoaldosteronism/Type 4 RTA) or receptor blockade causes hyperkalemia.

BOARD TRAP

WRONG to forget that hyperkalemia itself is a direct stimulus to aldosterone independent of the renin axis. Discriminator: a high K⁺ should drive aldosterone up directly; if it doesn't (Type 4 RTA, MRA use, Addison's), suspect an adrenal or receptor problem — check aldosterone and renin to localize the lesion.

5. Pillar 1: Aldo

WHAT YOU SEE

Principal cell cathedral, aldo binding MR

WHAT IT ENCODES

Pillar 1: aldosterone binds the cytosolic mineralocorticoid receptor in principal cells, increasing transcription of ENaC, Na/K-ATPase, and ROMK. The net effect is enhanced Na⁺ reabsorption and K⁺ secretion. This is the target of spironolactone/eplerenone (MR antagonists).

BOARD TRAP

WRONG to expect spironolactone to spare K⁺ only mildly. Discriminator: MR antagonists (spironolactone/eplerenone) directly block this pillar and are potent causes of hyperkalemia, especially in CHF/CKD or combined with ACEi/ARB — monitor K⁺ closely after starting or up-titrating.

6. Pillar 2: ENaC

WHAT YOU SEE

ENaC channel, Na⁺ entering

WHAT IT ENCODES

Pillar 2: the epithelial sodium channel (ENaC) on the apical principal-cell membrane lets luminal Na⁺ enter the cell, generating the lumen-negative transepithelial voltage that drives K⁺ secretion through ROMK. ENaC activity depends on aldosterone and on adequate luminal Na⁺. It is blocked by amiloride and triamterene.

BOARD TRAP

WRONG to assume amiloride/triamterene cause hyperkalemia by an aldosterone mechanism. Discriminator: these block ENaC DIRECTLY (downstream of the receptor), so they cause hyperkalemia even when aldosterone is normal/high — contrast with spironolactone, which works at the MR. Note Liddle syndrome = ENaC gain-of-function (HYPOkalemia).

7. Na/K-ATPase

WHAT YOU SEE

Basolateral Na/K-ATPase pump

WHAT IT ENCODES

The basolateral Na/K-ATPase pumps Na⁺ out of and K⁺ into the principal cell, keeping intracellular K⁺ high so K⁺ can flow down its gradient into the lumen via ROMK. It is the energy-dependent engine that loads the cell with K⁺ for secretion. Aldosterone upregulates it; digoxin and severe ischemia inhibit it.

BOARD TRAP

WRONG to overlook Na/K-ATPase inhibition as a mechanism of hyperkalemia. Discriminator: digoxin toxicity blocks the Na/K-ATPase body-wide, causing a transcellular shift of K⁺ OUT of cells — hyperkalemia in acute digoxin toxicity is a marker of severity and an indication for digoxin-specific Fab.

8. ROMK

WHAT YOU SEE

ROMK channel secreting K⁺

WHAT IT ENCODES

ROMK (renal outer medullary K⁺ channel) is the apical channel through which K⁺ exits the principal cell into the urine, driven by the lumen-negative voltage. Its activity rises with aldosterone, high distal flow, and high tubular K⁺. It is the final common conduit for regulated K⁺ secretion.

BOARD TRAP

WRONG to ignore the loop-of-Henle link to ROMK. Discriminator: ROMK in the thick ascending limb is essential for the NKCC2 cotransporter; loss-of-function (Bartter syndrome type 2) causes a loop-diuretic-like salt wasting with HYPOkalemia — distinguish channel location/function when reasoning about K⁺ direction.

9. Lumen-neg voltage

WHAT YOU SEE

Lumen-negative voltage arrow

WHAT IT ENCODES

ENaC-mediated Na⁺ reabsorption leaves the lumen electrically negative, and this lumen-negative voltage is the electrochemical force that pulls K⁺ (and H⁺) out of the cell into the urine. Without Na⁺ reabsorption there is no driving voltage, and K⁺ secretion stalls. This couples Na⁺ handling directly to K⁺ excretion.

BOARD TRAP

WRONG to expect normal K⁺ secretion when distal Na⁺ reabsorption is impaired. Discriminator: anything that reduces ENaC-driven Na⁺ reabsorption (low distal Na⁺, ENaC blockers, low aldosterone) collapses the lumen-negative voltage and the K⁺-secretory drive — explaining hyperkalemia in those states.

10. Pillar 3: distal Na⁺

WHAT YOU SEE

Incoming sodium river, Na⁺ > 25 mM

WHAT IT ENCODES

Pillar 3: adequate distal Na⁺ delivery (urine Na⁺ > ~25 mM) is needed to fuel ENaC and generate the lumen-negative voltage. In avid-reabsorption states the proximal nephron grabs Na⁺ first, starving the distal segment. Thiazides and loops increase distal Na⁺ delivery, which is partly why they promote K⁺ wasting.

BOARD TRAP

WRONG ANSWER: blaming hyperkalemia in CHF or cirrhosis purely on the kidney itself. Discriminator: in CHF, cirrhosis, and volume depletion, intense proximal Na⁺ reabsorption leaves little distal Na⁺ — no Na⁺ delivery means no ENaC activity and no K⁺ secretion, so K⁺ is retained despite (often high) aldosterone.

11. Pillar 4: urine flow

WHAT YOU SEE

Urine flow current sweeping K⁺ away

WHAT IT ENCODES

Pillar 4: brisk tubular flow continuously sweeps secreted K⁺ away from the apical membrane, maintaining a steep cell-to-lumen K⁺ gradient that sustains ongoing secretion. High flow also activates flow-sensitive maxi-K (BK) channels for additional K⁺ excretion. Low flow lets luminal K⁺ accumulate and shuts secretion down.

BOARD TRAP

WRONG to expect adequate K⁺ excretion in an oliguric patient even if aldosterone and Na⁺ delivery are fine. Discriminator: low urine flow (oliguria, dehydration) abolishes the gradient and halts K⁺ secretion — a key reason AKI/oliguria causes hyperkalemia and why restoring urine flow (fluids, loop diuretics) helps.

12. Urine flow > 40

WHAT YOU SEE

Banana river to bladder, normal > 40 mL/day

WHAT IT ENCODES

Normal urinary K⁺ excretion is roughly > 40-80 mM/day, reflecting balance with dietary intake of ~40-120 mM. A spot urine K⁺ that is inappropriately LOW (< ~20 mM) in the face of hyperkalemia points to a renal/tubular secretory defect rather than a load problem. Urine K⁺ helps separate renal from extrarenal causes.

BOARD TRAP

WRONG to assume hyperkalemia must be from intake when urine K⁺ is low. Discriminator: an inappropriately low urine K⁺ during hyperkalemia indicts impaired renal secretion (low aldosterone, ENaC block, low Na⁺/flow), whereas a high urine K⁺ with high serum K⁺ suggests an overwhelming load or shift with intact kidneys.

13. TTKG > 8

WHAT YOU SEE

TTKG > 8 output gauge

WHAT IT ENCODES

The transtubular potassium gradient (TTKG = [urine K⁺/plasma K⁺] / [urine osm/plasma osm]) estimates distal K⁺ secretory drive. In hyperkalemia, a TTKG > ~7-8 indicates an appropriate kidney response (intact aldosterone/secretion), implicating a load or shift. Validity requires urine osm > plasma osm and urine Na⁺ > 25.

BOARD TRAP

WRONG ANSWER: a low TTKG in hyperkalemia means the kidney is responding normally. Discriminator: in hyperkalemia an INAPPROPRIATELY LOW TTKG (< 5-7) signals a secretory defect — hypoaldosteronism (Type 4 RTA), MR/ENaC blockade — not a load; a fludrocortisone challenge that raises TTKG confirms aldosterone deficiency/resistance. (Note: TTKG is now considered less reliable but remains testable.)

14. 4-pillar scroll

WHAT YOU SEE

Four-pillar pearl scroll

WHAT IT ENCODES

Bottom line: renal K⁺ secretion needs aldosterone + ENaC + distal Na⁺ delivery + urine flow, and failure of ANY single pillar causes hyperkalemia. Map each clinical cause to its pillar — Addison's/Type 4 RTA (aldosterone), amiloride/triamterene (ENaC), CHF/cirrhosis/volume depletion (distal Na⁺), oliguria/AKI (flow). This framework turns a long hyperkalemia differential into four mechanistic buckets.

BOARD TRAP

WRONG to treat the four pillars as redundant backups for one another. Discriminator: they are serial, non-redundant requirements — a single failed pillar is sufficient to cause hyperkalemia, so identify WHICH pillar is broken to target therapy (e.g., fludrocortisone for aldosterone deficiency, fluids/diuretics to restore Na⁺ delivery and flow).